How To Use This Guide

Controlled flight by humans was attempted early in the last millenium but only mastered for heavier-than-air vehicles in the last century. Tremendous progress was achieved in the twentieth century in aircraft performance and mission capability through research in flight controls, aircraft stability, and propulsion. Modern technology was applied to aeronautics in the last century as aggressively as in other fields, such as medicine, communications, and geosciences. So many advances have been made, that today the demands of performance and maneuverability for many advanced aircraft designs require extensive use of computers to aid the pilot in controlling flight.

With some simple inexpensive materials, you can mount an exciting and productive unit for children that incorporates science, mathematics, and technology education. The many activities contained in this teaching guide emphasize hands-on involvement, prediction, data collection and interpretation, teamwork, and problem solving. The guide also contains background information about aeronautical research that can help students learn how airplanes fly.

Following the background sections are a series of activities that demonstrate the basic science of aeronautics while offering challenging tasks in design. Each activity employs basic and inexpensive materials. In each activity you will find construction diagrams, material and tools lists, and instructions. A brief background section within each activity elaborates on the concepts covered and points back to the introductory material in this guide. Also included is information about where the activity applies to science and mathematics standards, assessment ideas, and extensions.

Because many of the activities and demonstrations apply to more than one subject area, a matrix chart identifies

